

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

(192411015)

I M. Mokshitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Mokshitha

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10m
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	3	3	3	3	3
Decision Making	20%	2	1	1	1	1
Clarity of Response	15%	1	1	1	1	1

26/30

CO2-AT2 Analytical problems

Scenario-Based

Network Block: 192.168.100.0/24

- Total IP Addresses = 256

Subnet allocation:

- School of Computing $\rightarrow /25$
- School of Electronics $\rightarrow /26$
- School of Mechanical Engineering $\rightarrow /27$

Step 1: School of Computing (125):-

- Subnet Mask: 255.255.255.128
- Total Addresses = $2^{32-25} = 128$
- Usable Hosts = $128 - 2 = 126$
- Network address: 192.168.100.0
- Host Range: 192.168.100.1 - 192.168.100.126
- Broadcast Address: 192.168.100.127

Step 2: School of Electronics (126):-

- Subnet mask: 255.255.255.192
- Total Addresses: $2^{32-26} = 64$
- Usable Hosts = $64 - 2 = 62$
- Network Address: 192.168.100.128
- Host Range: 192.168.100.129 - 192.168.100.190
- Broadcast Address: 192.168.100.191

Step 3: School of Mechanical Engineering (127)

- Subnet Mask: 255.255.255.224

- Total Addresses: $2^{32-27} = 32$

- Usable Hosts = $32 - 2 = 30$

- Network Address: 192.168.100.192

- Host Range: 192.168.100.193 - 192.168.100.222

- Broadcast Address: 192.168.100.223

Step 4: Remaining unused Address Range

Addresses used:

- 192.168.100.0 - 192.168.100.223

Remaining addresses:

- Network Address: 192.168.100.224/27

- Host Range: 192.168.100.225 - 192.168.100.254

- Broadcast Address: 192.168.100.255

- Usable Hosts: 30

Given

- Network Block: 10.10.0.0/16

- Total IP Addresses = 65,536

Software Development (124)

Subnet Mask: 255.255.255.0

Total Addresses = $2^{32-34} = 256$

Usable Hosts = $256 - 2 = 254$

Networking Address: 10.10.0.0

Host range: 10.10.0.1 - 10.10.0.254

Broadcast Address: 10.10.0.255

Step 2: Quality Assurance (128)

Subnet Mask: 255.255.255.128

Total Address: $2^{32-25} = 128$

Usable Hosts = $128 - 2 = 126$

Network Address: 10.10.1.0

Hosts Range: 10.10.1.1 - 10.10.1.126

Broadcast Address: 10.10.1.127

Step 3: Human Resources (126)

Subnet Mask: 255.255.255.192

Total Address = $2^{32-26} = 64$

Usable Hosts = $64 - 2 = 62$

Usable Address = 10.10.1.128

3)

Given IPv6 Network.

2001:0db8:abcd:0000:0000:0000:0000/64

1) Convert to Compressed Notation

Remove leading zeros and replace consecutive 0000 groups

Compressed Address:

2001:db8:abcd:164

2) Expand the Compressed Address

Expand Address:

2001:0db8:abcd:0000:0000:0000:0000/64

3) Identify the Network prefix (164)

With a 164 prefix, the first 64 bits

Network prefix:

2001:0db8:abcd:0000::/64

4) Calculate the Total no. of host addresses

IPv6 uses 128-bit addresses.

$$\text{Host bits} = 128 - 64 = 64$$

$$\text{Total host} = 2^{64}$$

Given Routing table:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination Ip: 192.168.2.45

• Destination Subnet:

192.168.2.0/24

• Matching Routing Table

192.168.2.0/24

• Next Hop network/Interface:

192.168.2.0/24

• Default Route

0.0.0.0/0

5) Administration LAN

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Host Range: 192.168.10.1 - 192.168.10.62
- Three pc IPs: 192.168.10.2, 192.168.10.3,
192.168.10.4

• Default Gateway: 192.168.10.1

Finance LAN (192.168.10.64/26)

• Network address: 192.168.10.64

• Broadcast address: 192.168.10.127

• Host Range: 192.168.10.65 - 192.168.10.126

• Default Gateway: 192.168.10.65